Chain Link Security: Trends in Simultaneous HTTPS, ECH, and DNSSEC Use

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Goals & Motivation: Surveying Usage of HTTPS, ECH, and DNSSEC

- DNS HTTPS, ECH, and DNSSEC are relatively new and related web protocols.
  - ECH allows for the encryption of TLS handshakes, DNSSEC is a series of cryptographic signature protocols that ensure better integrity and authenticity of DNS records.

- It could be advantageous to use DNSSEC in conjunction with HTTPS to authenticate records and prevent forgery, as well as to use it with ECH to authenticate handshake keys.

- What will be the larger environment for deploying DNS HTTPS records with ECH and DNSSEC in 2026?

Goals & Motivation: The 10,000 Domain Case Study

- The previous studies of HTTPS that we draw from focus on very large data sets, where usually millions of domains are surveyed.
  - For both time purposes and to approach the data from a different angle, we'll be surveying the top 10,000 domains.
- Qualitatively, examining the top 10,000 domains is a different beast then examining the top 1,000,000, not just quantitatively. We cannot directly compare deployment values because of that.
- The infrastructure and purpose of a top 10 domain is different than that of the bottom 10 out of 10,000 domains.
  - Extending that, the bottom 100 out of 10,000 domain is also going to be different than the bottom 100 of 1,000,000 domains. They may have different motivations regarding their use of security protocols.
- Think of this as a case study of HTTPS, ECH, DNSSEC usage over the thin top layer of the 10,000 most visited domains of 2025.

Top and Bottom 15 of 10,000 Domains

```
1   {"domain": "google.com", "h
2   {"domain": "microsoft.com",
3   {"domain": "apple.com", "ha
4   {"domain": "facebook.com",
5   {"domain": "youtube.com", "
6   {"domain": "amazon.com", "h
7   {"domain": "bing.com", "has
8   {"domain": "yahoo.com", "ha
9   {"domain": "msn.com", "has_
10  {"domain": "live.com", "has
11  {"domain": "instagram.com",
12  {"domain": "linkedin.com",
13  {"domain": "netflix.com", "
14  {"domain": "spotify.com", "
15  {"domain": "office.com", "h
```

```
9986   {"domain": "dropmms.com", "ha
9987   {"domain": "flashscore.pt", "
9988   {"domain": "gov.uz", "has_htt
9989   {"domain": "hamariweb.com", "
9990   {"domain": "macour.jp", "has_
9991   {"domain": "manhwa-raw.com",
9992   {"domain": [redacted], "ha
9993   {"domain": [redacted],
9994   {"domain": "qan.az", "has_htt
9995   {"domain": [redacted], "
9996   {"domain": "webteb.com", "has
9997   {"domain": [redacted],
9998   {"domain": "nsdl.com", "has_h
9999   {"domain": "telegram.com", "h
10000  {"domain": "bereal.com", "has
```

Background: Hypertext Transfer Protocol Secure (HTTPS) & Transport Layer Security (TLS)

- HTTPS is a layer security protocol that’s used to send data between a web browser and website.

- HTTPS uses TLS to provide data confidentiality or authentication between the client and the server for services that require them.

- TLS is the cryptographic protocol that provides encryption and authentication for HTTPS.

Background: HTTPS Parameters & Dynamic Configs

- SvcPriority: The priority of the record relative to another record of the same name.
  - 0: AliasMode, which alias a domain name to the target domain.
  - Other values: ServiceMode, which provides information specific to a service endpoint
- SvcParams
  - alpn (Application-Layer Protocol Negotiation): specifies additional application protocols.
  - ipv4hint/ipv6hint: suggest IPv4/IPv6 addresses for reaching the endpoint.
- Dynamic Configurations
  - Occurs when at least one parameter uses an expression instead of a fixed value
  - <ftp:config name="ftp" **defaultWriteEncoding="#[attributes.queryParams.encoding]**">, <ftp:connection username="foo" password="mySecret" />, </ftp:connection>

Background: Encrypted Client Hello (ECH)

- Hides the identity of the target TLS server by encrypting the Server Name Indication (SNI) in the client Hello using the public key of the client-facing server.

- The client-facing server can decrypt the ECH extension before forwarding the client Hello to the correct server.

- Limits access to information that a particular user is visiting your website, ensuring that it is not unnecessarily shared with intermediaries.

Background: Domain Name System Security Extensions (DNSSEC)

- A suite of specifications designed to fortify DNS data with origin authentication, data integrity, and authenticated denial of existence

- Prevents DNS spoofing attacks by allowing authoritative name servers to digitally sign RRsets using public-key cryptography

- The DNSSEC resolver verifies that the information originates from a legitimate server and that the data matches the information on authoritative DNS servers by checking signatures in the DNSSEC servers' answers

Approach

- We would query the top 10,000 pay-level (no sub-domains) domains from over the course of 2025, as well as for each month of the year, using a list provided by the site Tranco.
- We fetch the information regarding the following Parameters:
  - HTTPS RR (Does the domain use HTTPS)
  - ECH Usage (Does the domain use HTTPS and ECH)
  - DNSSEC Usage (Does the Domain use HTTPS and DNSSEC)
  - SvcPriority Mode Used (Service or Alias Mode)
  - ALPN Usage
  - ipv4hint Usage
  - Ipv6hint Usage
  - Dynamic Configuration Usage
- We extracted and calculated the data of each list, but notably we only had the resources to extract the information of the date of extraction.
  - Our finalized extraction of data occurred on March 7th, 2026.

Approach Continued

- We queried each domain using the library DNSPython to ask them what their HTTPS Resource Record said.

- Additionally, we take that tally and calculate overall averages.

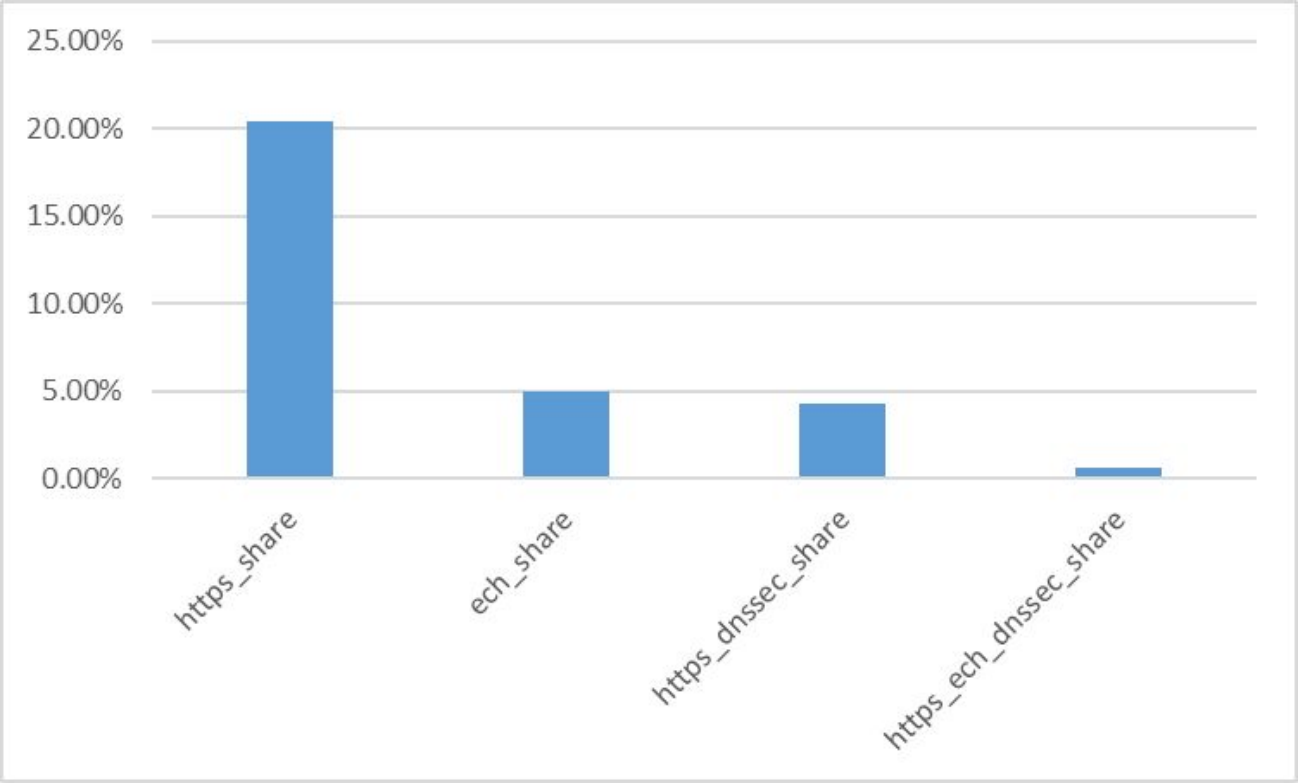
- For qualitative analysis, we took made lists from the extracted domains, organized by their protocol or parameter use (HTTPS + ECH, HTTPS + DNSSEC, etc.).
  - We then categorize them in two ways. First is site category, the broad category that describes a site, like a news site. Second is desired content, which describe the main type of content a user would seek from this site, like articles.

Data Calculation Results

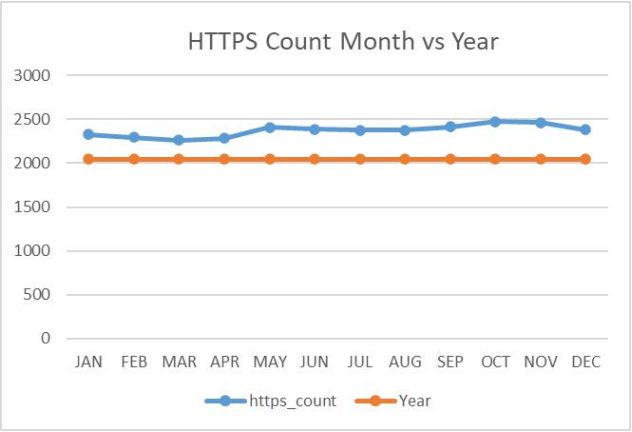
Domain/Protocol	HTTPS	HTTPS+ECH	HTTPS+DNSSEC	HTTPS+ECH+DNSSEC
Count	2044	497	430	59
Share(%)	20.44	4.97	4.30	0.59

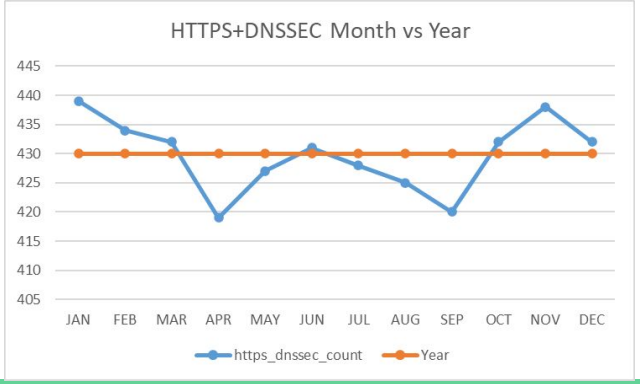
Domain/HTTPS params & dynamic config	Alias mode	Service mode	alpn	ivp4	ivp6	Dynamic config
Count	3	2041	2029	1977	1689	2037
Share(%) vs all domain	0.03	20.41	20.29	19.77	16.89	20.37
Share(%) vs https domain	0.14	99.85	99.27	96.72	82.63	99.66

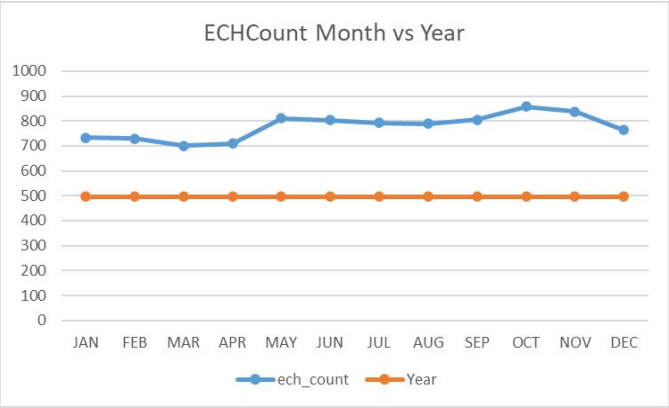
Simultaneous HTTPS, ECH, and DNSSEC usage

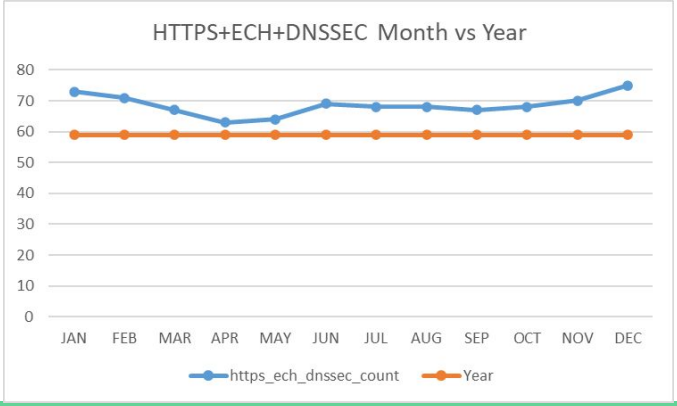


Stats on the Monthly Top 10,000 List









Speculation: Why is DNSSEC Usage So Low?

- Reading developer feedback, it seems that DNSSEC has a set of weaknesses [1]:
  - It's difficult to implement and maintain, with key rollovers that can cause outages for domains if not handled correctly.
  - The infrastructure to support it, like client-side validators to check signatures, were just not there.
  - DNSSEC mostly protects the server-side records, not client-side, which a lot of users want, and is mostly useful for high-security users, like banks.
  - There are attractive alternative security methods overshadow it, like DNS over HTTPS, which encrypts DNS queries. [2] [3]
  - And all that with the extra resource costs to support the cryptology system means that it's unattractive.

Speculation: Why is ECH Usage So Low?

- ECH requires coordinated support across DNS, TLS, and server infrastructure, so domains with less use cases are likely to disuse it to management load [4].
- Cloudflare encryption masks the domain in a TLS handshake, making content filtering and monitoring difficult, which affects things like firewalls [5].
- Cloudflare turns on ECH by default for free plans, with the ability to turn it on and off is exclusive to other plans [6].
- Studies that measures a higher number of domains usually have a higher amount of ECH domains. In 2022, ECH domains accounted for roughly 7.5% of TLS traffic in general, compared to the almost 5 % of HTTP and ECH domains in the top 10,000 [7].
    - Cloudflare is a major domain provider. Across a six month period from November 2023 to April 2024, the mean of domains saw a Cloudflare name server share of 98.89% [8].
    - It's likely that the top 10,000 domains, being more popular than the domains below it, are more likely to have paid Cloudflare plans that can turn off ECH, given their potential for having additional resources that go with their popularity.

Exploring the Ecosystem of DNS HTTPS Resource Records:
An End-to-End Perspective

Name Server (NS) Category	Dynamic Tranco top 1M		Overlapping	
	Mean (%)	Std.	Mean (%)	Std.
Full *Cloudflare* NS	99.89	0.03	99.87	0.04
None *Cloudflare* NS	0.11	0.03	0.13	0.04
Partial *Cloudflare* NS	< 0.01	-	< 0.01	-

Table 2: The average and standard deviation of the percentage of apex domains (with HTTPS records) served by *Cloudflare* vs non-*Cloudflare* name servers.

[8]

Why Do Particular Domains Use These Protocols

- To better understand this and ground our speculation, we did our qualitative analysis.
- As discussed before, we'll be looking at the raw extracted data we get from domains, and dividing the by the various values we're recording, like ECH usage. These domains are then given two category values, site category and desired content.
- We've begun examining these records, but currently the only available report is on the usage of HTTPS + ECH + DNSSEC.
- If possible, other records will be found in our report, due to both presentation and parsing time constraints.
- We can still expand on our tentative findings if needed.

HTTPS + ECH + DNSSEC Pivot Table

Count of Content Type	Desired Content											
Site Category	Adult_Entertainment	Articles	Commerce	Communication	Conditional_Software	Database_Info	File_Sharing	Pirated_Media	Text	User_Input_Output	Utility_Software	Grand Total
Chat_Site				5								5
Health_Site		1										1
Image_Browser	5							2				7
Interest_Site		3				1						4
News_Site		5										5
Ref									1			1
Service_Site	1		1		2							4
Shopping_Site			2									2
Software_Sharing					1			1			2	4
Text_Sharing							1	2	1			4
Utility_Site						2			1	2		5
Video_Site	17											17
Grand Total	23	9	3	5	3	3	1	5	3	2	2	59

HTTPS + ECH + DNSSEC Qualitative Analysis

- Adult Entertainment and Pirated Media are the two largest types of desired content in domains that use HTTPS + ECH + DNSSEC. The reasons probably have to do with ECH's ability to hide SNI. Adult entertainment can be a taboo subject, while pirated media sites of a dubious legality, so association is not ideal. That creates an incentive to hide that information from other. Furthermore these sites tend to have targets on their backs, creating another incentive to hide the SNI.
- Most categories describe domains that heavily use large media files: video sites, image browsers, articles with large images. That makes these very reliant on CDNs for performance, which in turn makes them targets for cache poisoning, the act of injecting malicious data into a cache to control it. DNSSEC's validator serves as protection against these type of attacks [9].

Conclusion

- As it turns out, HTTPS + ECH + DNSSEC is a difficult to implement suite of protocols. Unless they really directly benefit, most domains will avoid the implementing the two options in 2026, to reduce maintenance difficulty and server load. This is especially reflected in the top 10,000 domains, who are largely able turn ECH due to their resources.
- This is reflected in the small tallies for ECH and DNSSEC, especially with them all together.
- That being said, there are uses case for these protocols. DNSSEC sees use in heavy security situations like banks, while ECH sees use in privacy contingent situation. Domains in that latter situation, that also use many large media files, are themselves a niche for HTTPS + ECH + DNSSEC, which can provide security and confidentiality in these situations.

GitHub Repository

- Our data collection code, actual data, internal documents, and existing work are available at our GitHub Repository.
- https://github.com/RhoAI/CS578-Project-Chain-Link-Security-Data-Files

Citations

[1] “All your DNSSEC base are belong to US,” PwnDefend, https://www.pwndefend.com/2025/11/25/all-your-dnssec-base-are-belong-to-us/ (accessed Mar. 11, 2026).

[2] Chromium, https://issues.chromium.org/issues/40426170#comment23 (accessed Mar. 11, 2026).

[3] A. Kotylevskiy, “DNSSEC: Should you use it? an expert’s view on the pros, cons, and when to implement - netlas blog,” Netlas, https://netlas.io/blog/what_is_dnssec/#:~:text=DNSSEC%20does%20not%20protect%20against,responses%20by%20removing%20DNSSEC%20configuration (accessed Mar. 11, 2026).

[4]Khandkar, V. S., Hanawal, M. K., & Kulkarni, S. G. (2022). Challenges in Adapting ECH in TLS for Privacy Enhancement over the Internet. https://doi.org/10.48550/arxiv.2207.01841

[5] D. Cid, “Encrypted client hello (ECH): Impact on content filtering,” CleanBrowsing, https://cleanbrowsing.org/articles/how-encrypted-client-hello-ech-impacts-content-filtering-solutions-challenges-and-adaptations (accessed Mar. 11, 2026).

[6] “Ech Protocol,” Cloudflare Docs, https://developers.cloudflare.com/ssl/edge-certificates/ech/ (accessed Mar. 11, 2026).

[7] Martino Trevisan, Francesca Soro, Marco Mellia, Idilio Drago, and Ricardo Morla. 2023. Attacking DoH and ECH: Does Server Name Encryption Protect Users’ Privacy? ACM Trans. Internet Technol. 23, 1, Article 19 (February 2023), 22 pages. https://doi.org/10.1145/3570726

[8] Hongying Dong, Yizhe Zhang, Hyeonmin Lee, Shumon Huque, and Yixin Sun. 2024. Exploring the Ecosystem of DNS HTTPS Resource Records: An End-to-End Perspective. In Proceedings of the 2024 ACM on Internet Measurement Conference (IMC '24). Association for Computing Machinery, New York, NY, USA, 423–440.

[9] N. Gul, “How to avoid web cache poisoning attacks,” Snyk, https://snyk.io/blog/how-to-avoid-web-cache-poisoning-attacks/ (accessed Mar. 11, 2026).

Questions?

Thank You!